Practice Task 1 Demo File for Data Preparation and Cleaning

1. Before Cleaning

Date	Client	Contact	Department	Payment	Revenue	Profit	Profit Margin
5/30/2023	AMAZON.COM, INC. (XNAS:AMZN)	Bill SmITH	Cloud Tech_Texas	Transfer	$4,500	$598	13.3%
5/30/2023	TESLA, INC. (XNAS:TSLA)	KEN Singh	Strategy_New York	PayPal	$3,800	$1,045	27.5%
5/30/2023	NETFLIX, INC. (XNAS:NFLX)	Harley Fritz	Strategy_New York		$3,713	$1,009	27.2%
5/30/2023	THE PROCTER & GAMBLE COMPANY (XNYS:PG)	Nyla Novak	Operations_Florida			$779	#DIV/0!
5/30/2023	THE GOLDMAN SACHS GROUP, INC. (XNYS:GS)	David Rasmussen	Operations_Florida	Check	$5,000	$684	13.7%
5/31/2023	JPMORGAN CHASE & CO. (XNYS:JPM)	IVAN HINEY	Cloud Tech_Texas	Transfer	$6,100	$544	8.9%
5/31/2023	MORGAN STANLEY (XNYS:MS)	JONha Ma	Cloud Tech_Texas	Transfer	$4,625	$670	14.5%
5/31/2023	CITIGROUP INC. (XNYS:C)	Jordan Boone	Cloud Tech_Texas	Transfer	$3,800	$2,045	53.8%
5/31/2023	BANK OF AMERICA CORPORATION (XNYS:BAC)	Kylee Townsend	Cloud Tech_Texas	Card	$3,600	$1,564	43.4%
5/31/2023	WALMART INC. (XNYS:WMT)	Nora Rollins	Cloud Tech_Texas	Check	$5,100	$1,220	23.9%
5/31/2023	TARGET CORPORATION (XNYS:TGT)	BRENDAN Wallace	Cloud Tech_Texas	Check	$4,750	$1,435	30.2%
5/31/2023	COSTCO WHOLESALE CORPORATION (XNAS:COST)	Conor Wise	Operations_Florida	Transfer	$6,000	$998	16.6%
5/31/2023	MCDONALD'S CORPORATION (XNYS:MCD)	Steven MIChael	Big Data_California	Check	$4,500	$780	17.3%
6/1/2023	EXXON MOBIL CORPORATION (XNYS:XOM)	Lucia Mckay	Big Data_California	Card		$1,044	#DIV/0!
6/1/2023	VERIZON COMMUNICATIONS INC. (XNYS:VZ)	JoSE Roach	Big Data_California	Transfer	$3,713	$1,222	32.9%
6/1/2023	THE HOME DEPOT, INC. (XNYS:HD)	Franklin WRIGT	Big Data_California	Transfer	$4,950	$1,065	21.5%
6/1/2023	CISCO SYSTEMS, INC. (XNAS:CSCO)	Alia Thornton	Operations_Florida	Transfer	$4,750	$810	17.1%
6/1/2023	CHEVRON CORPORATION (XNYS:CVX)	Denzel Flores	Operations_Florida	Transfer	$7,320	$933	12.7%
5/31/2023	COSTCO WHOLESALE CORPORATION (XNAS:COST)	Conor Wise	Operations_Florida	Transfer	$6,000	$998	16.6%
5/31/2023	MCDONALD'S CORPORATION (XNYS:MCD)	Steven MIChael	Big Data_California	Check	$4,500	$780	17.3%
6/1/2023	AT&T INC. (XNYS:T)	Bruno CordOVA	Big Data_California	Transfer	$5,088	$655	12.9%
6/1/2023	INTEL CORPORATION (XNAS:INTC)	jaylynn napp	Big Data_California	Transfer	$4,500	$722	16.0%
6/1/2023	GENERAL MOTORS COMPANY (XNYS:GM)	Bruce RICH	Big Data_California	Card	$4,250	$901	21.2%
6/2/2023	MICROSOFT CORPORATION (XNAS:MSFT)	Arturo Moore	Big Data_California	PayPal	$5,250	$1,349	25.7%
6/2/2023	COMCAST CORPORATION (XNAS:CMCSA)	Bryce Carpenter	Strategy_New York	PayPal	$6,500	$1,288	19.8%
6/2/2023	DELL TECHNOLOGIES INC. (XNYS:DELL)	Jaidyn ANDERSEN	Strategy_New York	PayPal	$7,500	$1,664	22.2%
6/2/2023	JOHNSON & JOHNSON (XNYS:JNJ)	MARK WALM	Strategy_New York	Transfer	$5,500	$1,320	24.0%
6/2/2023	FEDEX CORPORATION (XNYS:FDX)	HARRY LEE	Strategy_New York	Transfer	$4,625	$1,001	21.6%
6/2/2023	GENERAL ELECTRIC COMPANY (XNYS:GE)	JOSH JOHNSON	Strategy_New York	Transfer	$4,500	$960	21.3%
6/2/2023	LOCKHEED MARTIN CORPORATION (XNYS:LMT)	Mik Naam	Strategy_New York	Card	$5,400	$540	10.0%
5/30/2023	THE GOLDMAN SACHS GROUP, INC. (XNYS:GS)	David Rasmussen	Operations_Florida	Check	$5,000	$684	13.7%

a. Data Cleaning Start | Data Cleaning Start (2) | Client_dim | State_dim | Payment_dim | Department_dim | FactTable | VLookUp

2. After Cleaning

a.

A	B	C	D	E	F	G	H	I	J
trans_id	Date	client_id	Contact	department_id	state_id	payment_id	Revenue	Profit	Profit Margin
t-101	5/30/2023	AMAZON.COM, INC.	Bill Smith	Cloud Tech	Texas	Transfer	$ 4,500.00	$ 598.00	13.3%
t-102	5/30/2023	TESLA, INC.	Ken Singh	Strategy	New York	PayPal	$ 3,800.00	$ 1,045.00	27.5%
t-103	5/30/2023	NETFLIX, INC.	Harley Fritz	Strategy	New York	Others	$ 3,712.50	$ 1,009.00	13.7%
t-104	5/30/2023	THE PROCTER & GAMBLE COMPANY	Nyla Novak	Operations	Florida	Others	$ 4,994.00	$ 779.00	15.6%
t-105	5/30/2023	THE GOLDMAN SACHS GROUP, INC.	David Rasmussen	Operations	Florida	Check	$ 5,000.00	$ 684.00	13.7%
t-106	5/31/2023	JPMORGAN CHASE & CO.	Ivan Hiney	Cloud Tech	Texas	Transfer	$ 6,100.00	$ 544.00	8.9%
t-107	5/31/2023	MORGAN STANLEY	Jonha Ma	Cloud Tech	Texas	Transfer	$ 4,625.00	$ 670.00	14.5%
t-108	5/31/2023	CITIGROUP INC.	Jordan Boone	Cloud Tech	Texas	Transfer	$ 3,800.00	$ 2,045.00	53.8%
t-109	5/31/2023	BANK OF AMERICA CORPORATION	Kylee Townsend	Cloud Tech	Texas	Card	$ 3,600.00	$ 1,564.00	43.4%
t-110	5/31/2023	WALMART INC.	Nora Rollins	Cloud Tech	Texas	Check	$ 5,100.00	$ 1,220.00	23.9%
t-111	5/31/2023	TARGET CORPORATION	Brendan Wallace	Cloud Tech	Texas	Check	$ 4,750.00	$ 1,435.00	30.2%
t-112	5/31/2023	COSTCO WHOLESALE CORPORATION	Conor Wise	Operations	Florida	Transfer	$ 6,000.00	$ 998.00	16.6%
t-113	5/31/2023	MCDONALD'S CORPORATION	Steven Michael	Big Data	California	Check	$ 4,500.00	$ 780.00	17.3%
t-114	6/1/2023	EXXON MOBIL CORPORATION	Lucia Mckay	Big Data	California	Card	$ 4,994.00	$ 1,044.00	20.9%
t-115	6/1/2023	VERIZON COMMUNICATIONS INC.	Jose Roach	Big Data	California	Transfer	$ 3,712.50	$ 1,222.00	32.9%
t-116	6/1/2023	THE HOME DEPOT, INC.	Franklin Wrigt	Big Data	California	Transfer	$ 4,950.00	$ 1,065.00	21.5%
t-117	6/1/2023	CISCO SYSTEMS, INC.	Alia Thornton	Operations	Florida	Transfer	$ 4,750.00	$ 810.00	17.1%
t-118	6/1/2023	CHEVRON CORPORATION	Denzel Flores	Operations	Florida	Transfer	$ 7,320.00	$ 933.00	12.7%
t-119	6/1/2023	AT&T INC.	Bruno Cordova	Big Data	California	Transfer	$ 5,087.50	$ 655.00	12.9%
t-120	6/1/2023	INTEL CORPORATION	Jaylynn Napp	Big Data	California	Transfer	$ 4,500.00	$ 722.00	16.0%
t-121	6/1/2023	GENERAL MOTORS COMPANY	Bruce Rich	Big Data	California	Card	$ 4,250.00	$ 901.00	21.2%
t-122	6/2/2023	MICROSOFT CORPORATION	Arturo Moore	Big Data	California	PayPal	$ 5,250.00	$ 1,349.00	25.7%
t-123	6/2/2023	COMCAST CORPORATION	Bryce Carpenter	Strategy	New York	PayPal	$ 6,500.00	$ 1,288.00	19.8%
t-124	6/2/2023	DELL TECHNOLOGIES INC.	Jaidyn Andersen	Strategy	New York	PayPal	$ 7,500.00	$ 1,664.00	22.2%
t-125	6/2/2023	JOHNSON & JOHNSON	Mark Walm	Strategy	New York	Transfer	$ 5,500.00	$ 1,320.00	24.0%
t-126	6/2/2023	FEDEX CORPORATION	Harry Lee	Strategy	New York	Transfer	$ 4,625.00	$ 1,001.00	21.6%
t-127	6/2/2023	GENERAL ELECTRIC COMPANY	Josh Johnson	Strategy	New York	Transfer	$ 4,500.00	$ 960.00	21.3%
t-128	6/2/2023	LOCKHEED MARTIN CORPORATION	Mik Naam	Strategy	New York	Card	$ 5,400.00	$ 540.00	10.0%
							$4,976		

3. Creating Dim Table

a.

A	B	C
Client	client_id	Contact
AMAZON.COM, INC.	c-101	Bill Smith
AT&T INC.	c-102	Bruno Cordova
BANK OF AMERICA CORPORATION	c-103	Kylee Townsend
CHEVRON CORPORATION	c-104	Denzel Flores
CISCO SYSTEMS, INC.	c-105	Alia Thornton
CITIGROUP INC.	c-106	Jordan Boone
COMCAST CORPORATION	c-107	Bryce Carpenter
COSTCO WHOLESALE CORPORATION	c-108	Conor Wise
DELL TECHNOLOGIES INC.	c-109	Jaidyn Andersen
EXXON MOBIL CORPORATION	c-110	Lucia Mckay
FEDEX CORPORATION	c-111	Harry Lee
GENERAL ELECTRIC COMPANY	c-112	Josh Johnson
GENERAL MOTORS COMPANY	c-113	Bruce Rich
INTEL CORPORATION	c-114	Jaylynn Napp
JOHNSON & JOHNSON	c-115	Mark Walm
JPMORGAN CHASE & CO.	c-116	Ivan Hiney
LOCKHEED MARTIN CORPORATION	c-117	Mik Naam
MCDONALD'S CORPORATION	c-118	Steven Michael
MICROSOFT CORPORATION	c-119	Arturo Moore
MORGAN STANLEY	c-120	Jonha Ma
NETFLIX, INC.	c-121	Harley Fritz
TARGET CORPORATION	c-122	Brendan Wallace
TESLA, INC.	c-123	Ken Singh
THE GOLDMAN SACHS GROUP, INC.	c-124	David Rasmussen
THE HOME DEPOT, INC.	c-125	Franklin Wrigt
THE PROCTER & GAMBLE COMPANY	c-126	Nyla Novak
VERIZON COMMUNICATIONS INC.	c-127	Jose Roach
WALMART INC.	c-128	Nora Rollins
CCA	c-129	Wesly Cunanan

Data Cleaning Start | Data Cleaning Start (2) | Client_dim | State_dim | Payment_dim | Department_dim | FactTable | VLookUp

b.

States	state_id
California	s-101
Florida	s-102
New York	s-103
Texas	s-104

Data Cleaning Start | Data Cleaning Start (2) | Client_dim | State_dim | Payment_dim | Department_dim | FactTable | VLookUp

c.

Payment	payment_id
Transfer	p-101
Card	p-102
Check	p-103
Others	p-104
PayPal	p-105

Data Cleaning Start | Data Cleaning Start (2) | Client_dim | State_dim | Payment_dim | Department_dim | FactTable | VLookUp

d.

Department	department_id
Big Data	d-101
Cloud Tech	d-102
Operations	d-103
Strategy	d-104

Data Cleaning Start | Data Cleaning Start (2) | Client_dim | State_dim | Payment_dim | Department_dim | FactTable | VLookUp

4. Creating the Fact Table

a.

trans_id	Date	client_id	department_id	state_id	Payment_ID	Payment	Revenue	Profit	Profit Margin
t-101	5/30/2023	c-101	d-102	s-104	p-101	Transfer	$ 4,500.00	$ 598.00	13.3%
t-102	5/30/2023	c-123	d-104	s-103	p-105	PayPal	$ 3,800.00	$ 1,045.00	27.5%
t-103	5/30/2023	c-121	d-104	s-103	p-104	Others	$ 3,712.50	$ 1,009.00	13.7%
t-104	5/30/2023	c-126	d-103	s-102	p-104	Others	$ 4,994.00	$ 779.00	15.6%
t-105	5/30/2023	c-124	d-103	s-102	p-103	Check	$ 5,000.00	$ 684.00	13.7%
t-106	5/31/2023	c-116	d-102	s-104	p-101	Transfer	$ 6,100.00	$ 544.00	8.9%
t-107	5/31/2023	c-120	d-102	s-104	p-101	Transfer	$ 4,625.00	$ 670.00	14.5%
t-108	5/31/2023	c-106	d-102	s-104	p-101	Transfer	$ 3,800.00	$ 2,045.00	53.8%
t-109	5/31/2023	c-103	d-102	s-104	p-102	Card	$ 3,600.00	$ 1,564.00	43.4%
t-110	5/31/2023	c-128	d-102	s-104	p-103	Check	$ 5,100.00	$ 1,220.00	23.9%
t-111	5/31/2023	c-122	d-102	s-104	p-103	Check	$ 4,750.00	$ 1,435.00	30.2%
t-112	5/31/2023	c-108	d-103	s-102	p-101	Transfer	$ 6,000.00	$ 998.00	16.6%
t-113	5/31/2023	c-118	d-101	s-101	p-103	Check	$ 4,500.00	$ 780.00	17.3%
t-114	6/1/2023	c-110	d-101	s-101	p-102	Card	$ 4,994.00	$ 1,044.00	20.9%
t-115	6/1/2023	c-127	d-101	s-101	p-101	Transfer	$ 3,712.50	$ 1,222.00	32.9%
t-116	6/1/2023	c-125	d-101	s-101	p-101	Transfer	$ 4,950.00	$ 1,065.00	21.5%
t-117	6/1/2023	c-105	d-103	s-102	p-101	Transfer	$ 4,750.00	$ 810.00	17.1%
t-118	6/1/2023	c-104	d-103	s-102	p-101	Transfer	$ 7,320.00	$ 933.00	12.7%
t-119	6/1/2023	c-102	d-101	s-101	p-101	Transfer	$ 5,087.50	$ 655.00	12.9%
t-120	6/1/2023	c-114	d-101	s-101	p-101	Transfer	$ 4,500.00	$ 722.00	16.0%
t-121	6/1/2023	c-113	d-101	s-101	p-102	Card	$ 4,250.00	$ 901.00	21.2%
t-122	6/2/2023	c-119	d-101	s-101	p-105	PayPal	$ 5,250.00		0.0%
t-123	6/2/2023	c-107	d-104	s-103	p-105	PayPal	$ 6,500.00	$ 1,288.00	19.8%
t-124	6/2/2023	c-109	d-104	s-103	p-105	PayPal	$ 7,500.00	$ 1,664.00	22.2%
t-125	6/2/2023	c-115	d-104	s-103	p-101	Transfer	$ 5,500.00	$ 1,320.00	24.0%
t-126	6/2/2023	c-111	d-104	s-103	p-101	Transfer	$ 4,625.00	$ 1,001.00	21.6%
t-127	6/2/2023	c-112	d-104	s-103	p-101	Transfer	$ 4,500.00	$ 960.00	21.3%
t-128	6/2/2023	c-117	d-104	s-103	p-102	Card	$ 5,400.00	$ 540.00	10.0%
t-129	6/2/2023	c-117	d-104	s-103	p-102	Transfer	$ 5,400.00	$ 722.00	13.4%

Data Cleaning Start | Data Cleaning Start (2) | Client_dim | State_dim | Payment_dim | Department_dim | FactTable | VLookUp

5. VlookUp

a.

trans_id	Date	client_id	Client	Contact	department_id	Department	state_id	State	Payment_ID	Payment	Revenue	Profit	Profit Margin
t-101	5/30/2023	c-101	AMAZON.COM, INC.	Bill Smith	d-102	Cloud Tech	s-104	Texas	p-101	Transfer	$ 4,500.00	$ 598.00	13.3%
t-102	5/30/2023	c-123	TESLA, INC.	Ken Singh	d-104	Strategy	s-103	New York	p-105	PayPal	$ 3,800.00	$ 1,045.00	27.5%
t-103	5/30/2023	c-121	NETFLIX, INC.	Harley Fritz	d-104	Strategy	s-103	New York	p-104	Others	$ 3,712.50	$ 1,009.00	13.7%
t-104	5/30/2023	c-126	THE PROCTER & GAMBLE COMPANY	Nyla Novak	d-103	Operations	s-102	Florida	p-104	Others	$ 4,994.00	$ 779.00	15.6%
t-105	5/30/2023	c-124	THE GOLDMAN SACHS GROUP, INC.	David Rasmussen	d-103	Operations	s-102	Florida	p-103	Check	$ 5,000.00	$ 684.00	13.7%
t-106	5/31/2023	c-116	JPMORGAN CHASE & CO.	Ivan Hiney	d-102	Cloud Tech	s-104	Texas	p-101	Transfer	$ 6,100.00	$ 544.00	8.9%
t-107	5/31/2023	c-120	MORGAN STANLEY	Jonha Ma	d-102	Cloud Tech	s-104	Texas	p-101	Transfer	$ 4,625.00	$ 670.00	14.5%
t-108	5/31/2023	c-106	CITIGROUP INC.	Jordan Boone	d-102	Cloud Tech	s-104	Texas	p-101	Transfer	$ 3,800.00	$ 2,045.00	53.8%
t-109	5/31/2023	c-103	BANK OF AMERICA CORPORATION	Kylee Townsend	d-102	Cloud Tech	s-104	Texas	p-102	Card	$ 3,600.00	$ 1,564.00	43.4%
t-110	5/31/2023	c-128	WALMART INC.	Nora Rollins	d-102	Cloud Tech	s-104	Texas	p-103	Check	$ 5,100.00	$ 1,220.00	23.9%
t-111	5/31/2023	c-122	TARGET CORPORATION	Brendan Wallace	d-102	Cloud Tech	s-104	Texas	p-103	Check	$ 4,750.00	$ 1,435.00	30.2%
t-112	5/31/2023	c-108	COSTCO WHOLESALE CORPORATION	Conor Wise	d-103	Operations	s-102	Florida	p-101	Transfer	$ 6,000.00	$ 998.00	16.6%
t-113	5/31/2023	c-118	MCDONALD'S CORPORATION	Steven Michael	d-101	Big Data	s-101	California	p-103	Check	$ 4,500.00	$ 780.00	17.3%
t-114	6/1/2023	c-110	EXXON MOBIL CORPORATION	Lucia Mckay	d-101	Big Data	s-101	California	p-102	Card	$ 4,994.00	$ 1,044.00	20.9%
t-115	6/1/2023	c-127	VERIZON COMMUNICATIONS INC.	Jose Roach	d-101	Big Data	s-101	California	p-101	Transfer	$ 3,712.50	$ 1,222.00	32.9%
t-116	6/1/2023	c-125	THE HOME DEPOT, INC.	Franklin Wright	d-101	Big Data	s-101	California	p-101	Transfer	$ 4,950.00	$ 1,065.00	21.5%
t-117	6/1/2023	c-105	CISCO SYSTEMS, INC.	Alia Thornton	d-103	Operations	s-102	Florida	p-101	Transfer	$ 4,750.00	$ 810.00	17.1%
t-118	6/1/2023	c-104	CHEVRON CORPORATION	Denzel Flores	d-103	Operations	s-102	Florida	p-101	Transfer	$ 7,320.00	$ 933.00	12.7%
t-119	6/1/2023	c-102	AT&T INC.	Bruno Cordova	d-101	Big Data	s-101	California	p-101	Transfer	$ 5,087.50	$ 655.00	12.9%
t-120	6/1/2023	c-114	INTEL CORPORATION	Jaylynn Napp	d-101	Big Data	s-101	California	p-101	Transfer	$ 4,500.00	$ 722.00	16.0%
t-121	6/1/2023	c-113	GENERAL MOTORS COMPANY	Bruce Rich	d-101	Big Data	s-101	California	p-102	Card	$ 4,250.00	$ 901.00	21.2%
t-122	6/2/2023	c-119	MICROSOFT CORPORATION	Arturo Moore	d-101	Big Data	s-101	California	p-105	PayPal	$ 5,250.00	$ 1,349.00	25.7%
t-123	6/2/2023	c-107	COMCAST CORPORATION	Bryce Carpenter	d-104	Strategy	s-103	New York	p-105	PayPal	$ 6,500.00	$ 1,288.00	19.8%
t-124	6/2/2023	c-109	DELL TECHNOLOGIES INC.	Jaidyn Andersen	d-104	Strategy	s-103	New York	p-105	PayPal	$ 7,500.00	$ 1,664.00	22.2%
t-125	6/2/2023	c-115	JOHNSON & JOHNSON	Mark Walm	d-104	Strategy	s-103	New York	p-101	Transfer	$ 5,500.00	$ 1,320.00	24.0%
t-126	6/2/2023	c-111	FEDEX CORPORATION	Harry Lee	d-104	Strategy	s-103	New York	p-101	Transfer	$ 4,625.00	$ 1,001.00	21.6%
t-127	6/2/2023	c-112	GENERAL ELECTRIC COMPANY	Josh Johnson	d-104	Strategy	s-103	New York	p-101	Transfer	$ 4,500.00	$ 960.00	21.3%
t-128	6/2/2023	c-117	LOCKHEED MARTIN CORPORATION	Mik Naam	d-104	Strategy	s-103	New York	p-102	Card	$ 5,400.00	$ 540.00	10.0%
											$4,976		

Data Cleaning Start | Data Cleaning Start (2) | Client_dim | State_dim | Payment_dim | Department_dim | FactTable | VLookUp